

ANIMAL BIRTHDAY PREDICTOR (ABP)

MILESTONE 2: DELIVERABLES & TECHNICAL PROGRESS REPORT

Project: Animal Birthday Predictor Mobile App	Milestone Start Date: 20 August 2026
Milestone Scope: Equine Pregnancy, Scans & Care Module	Milestone Status: 100% COMPLETED (21/21 Deliverables)

Milestone Executive Summary: Milestone 2 encompassed the complete architecture, implementation, database synchronization, and end-to-end testing of the **Equine Pregnancy & Gestation Engine**, **Ultrasound Scans Module**, **Preventative Care Protocol**, **Recipient Mare Workflow**, and **Native Mobile Utilities**. All 21 specified deliverables have been successfully implemented and verified.

1. Milestone 2 Deliverables Checklist & Technical Notes

#	Deliverable	Status	Implementation Details & Technical Notes
1	Pregnancy Module	DONE (■)	Full equine gestation module supporting Natural, Chilled Semen, Frozen Semen, & ICSI/Embryo Transfer.
2	Pregnancy Details	DONE (■)	Live circular progress ring, remaining days countdown, and trimester breakdown indicator.
3	Three Pregnancy Scans	DONE (■)	Automated scheduling and tracking for Scan 1 (14–16d), Scan 2 (30d Heartbeat), and Scan 3 (45d Organogenesis).
4	Advanced Pregnancy Info	DONE (■)	Biological milestone timeline detailing embryonic development, physiological changes, and care tips.
5	Caslick Information	DONE (■)	Dedicated Caslick procedure tracking, surgical history logging, and pre-foaling opening reminders.
6	Fetal Sex	DONE (■)	Fetal gender recording (Colt, Filly, Unknown) captured and persisted during veterinary ultrasound scans.
7	Preventative Care	DONE (■)	Comprehensive preventative health manager for mares, foals, and canines with historical logs and reminders.
8	Vaccination Module	DONE (■)	Equine vaccine scheduler: Pneumabort-K at 5, 7, 9 months; Tetanus/Flu boosters at 10 months; Deworming cycles.
9	Dentist Records	DONE (■)	Equine dental checkup logging, tooth floating records, veterinarian notes, and recurring exam alerts.
10	Farrier Records	DONE (■)	Hoof trimming, corrective shoeing log, practitioner details, and 6–8 week recurring care schedule.
11	Phone Dial Integration	DONE (■)	AppPhoneLauncher integrated with native telephone dialer (`tel:`) and non-mobile clipboard fallback.
12	Record Detail Menu	DONE (■)	Contextual bottom sheets and popup menus for quick actions (Edit, View Profile, Log Scan, Add Care).
13	Complete Record Detail View	DONE (■)	Consolidated mare/foal profile screen displaying pedigree, chip ID, breeding history, scans, and care timeline.
14	Edit Existing Records	DONE (■)	Full update capability across animal profiles, breeding entries, dates, and medical notes.
15	Pregnancy Scan Editing	DONE (■)	Real-time editing of scan dates, confirmation toggles, ultrasound attachments, and vet comments.
16	Persist Scan Confirmations	DONE (■)	Permanent database persistence for Scans 1, 2, and 3 confirmation flags across application lifecycles.
17	Recipient Photo Saving	DONE (■)	Dedicated photo capture and cloud/local storage integration for recipient/surrogate mares.
18	Android Image Rendering Fix	DONE (■)	Resolved memory handling and aspect ratio rendering issues for camera/gallery image previews on Android.

19	Multiple Images Support	DONE (■)	Multi-photo management for animal avatars, ultrasound scans, and identification markings.
20	Complete CRUD Operations	DONE (■)	Full Create, Read, Update, and Delete operations verified across Animals, Pregnancies, Logs, & Contacts.
21	Testing of Horse Workflow	DONE (■)	End-to-end verification of complete equine breeding lifecycle on live device environments.

2. Deep Technical Breakdown by Architecture Layer

A. Equine Breeding & Calculation Engine

- **Mathematical Precision:** Implemented exact equine gestation periods based on ABP Section 4 specifications: Natural Cover (341 days), Chilled Semen (341 days), Frozen Semen (340 days), and ICSI / Embryo Transfer (332 days from transfer date / 340 days from donor ovulation).
- **Dual Mare Architecture:** Fully segregated genetic Dam from Recipient/Surrogate Mare to ensure accurate pedigree recording while calculating gestation milestones against the recipient's transfer dates.

B. Ultrasound Scans & Milestone Persistence

- **Scan 1 (Day 14–16):** Confirms embryonic vesicle and detects twins prior to fixation.
- **Scan 2 (Day 30):** Confirms positive heartbeat and checks viability.
- **Scan 3 (Day 45):** Verifies complete organogenesis and endometrial cup formation before wintering.
- **Data Persistence:** Confirmation toggles, scan images, attending vet notes, and fetal sex entries persist permanently in Supabase database tables.

C. Preventative Care & Health Management

- **Vaccination Protocol:** Automated tracking of critical anti-abortion vaccines (Pneumabort-K at months 5, 7, and 9) and pre-foaling passive immunity booster (Tetanus/Flu at month 10).
- **Dental & Hoof Care:** Historical logging and 6–8 week recurring alerts for Farrier trims and Equine Dentist floating.
- **Caslick Surgery Tracker:** Flagging mares with Caslick sutures with opening alerts 2–4 weeks prior to expected foaling.

D. Mobile Device & Platform Optimizations

- **Android Image Rendering Fix:** Implemented downsampling and caching handlers to prevent out-of-memory crashes on Android devices when taking high-res camera photos.
- **AppPhoneLauncher:** One-touch native telephone dialer (``tel:``) and email composer (``mailto:``) with clipboard fallback for devices without cellular capabilities.
- **Full CRUD Interface:** Contextual bottom sheets and edit screens enabling full record management.

MILESTONE 2 COMPLETION SIGN-OFF

Scope Status: All 21 Deliverables built, integrated, and verified.
Code Quality: Dart static analysis passed with zero errors.
Ready For: Stage 3 / Milestone 3 (45-Day Scan Certificates, Stud Foaling Diary, Twin Warnings & Advanced Celebratory UI).